Joobin Youn

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EDUCATION

Hanyang University | Seoul, South Korea **Mar. 2014 - Feb. 2020**

B.S. in Information Systems | GPA: 4.33/4.5

- Coursework: Probability and Statistics, Data Structures, Algorithms, Computer Architecture, Operating Systems, Computer Networks, Information Security, Database Systems, Object-Oriented Programming
- Course Paper: "Clipboard Hijacking Attacks and Vulnerability Analysis in Current Financial Transaction Systems" — Implemented and tested clipboard hijacking attacks across 4 browsers, 4 cryptocurrency wallets, and 3 anti-virus solutions; proposed virtual keyboard as a defensive countermeasure. Published on Medium · Demo on YouTube
- Honors: Summa Cum Laude (Feb. 2020)**,** Academic Excellence Award (Sep. 2018)**,** Dean's List — 7 semesters (2014, 2017–2019)
- Scholarships: Merit-Based & Career-Based Scholarships — 8 semesters (2014, 2017–2019)

University of Konstanz | Konstanz, Germany **Sep. 2018 - Feb. 2019**

International Exchange Program

- Coursework: Machine Learning, Data Mining
- Course Paper: "Right Whale Recognition" — Built a whale-head detector and compared HOG/SURF + linear SVM classifiers against CNNs in MATLAB for identifying individual whales from a highly imbalanced dataset.

PROFESSIONAL EXPERIENCE

Koscom | Seoul, South Korea | Korea's national capital markets IT infrastructure provider **Dec. 2019 - Present**

Information Security Engineer, Computer Emergency Response Team (CERT) **Nov. 2021 - Present**

- Designed and built AI-CERT, an LLM-powered SOC automation platform, by engineering a FastAPI relay server that bridges Splunk Enterprise Security with an on-premises LLM in a network-separated environment — automating 24/7 triage of ~10,000 low-severity events per day (~75% of total volume), only some of which were previously reviewed.
- As one of the core engineers, architected and deployed an enterprise-scale SIEM platform using Splunk, consolidating logs from 1,200+ heterogeneous security devices and internal solutions across Korea's primary financial network into a unified monitoring infrastructure; recognized with the Koscom Merit Award (Sep. 2022).
- Applied Splunk MLTK (unsupervised learning) to develop an anomaly-based detection policy for outbound traffic — clustering baseline log volumes to surface statistically significant deviations and replacing manual log review with automated detection.
- Built a pilot host firewall management system to test micro-segmentation in a Zero Trust testbed.
- Developed an enterprise IT Asset Management & Rogue Device Detection System, automating identification and tracking of unauthorized devices.

Information Security Engineer, Information Security Operations Team **Dec. 2019 - Nov. 2021**

- Led the redesign and migration of the enterprise IDS/IPS network, enhancing intrusion prevention capabilities for critical financial infrastructure.
- Established a secure IPSec VPN infrastructure connecting 30+ financial institutions, facilitating Koscom's role as a designated relay hub for Korea's national MyData ecosystem.
- Supported the design of secure cloud-native architectures for new service deployments and hybrid-cloud integrations, aligning with regulatory requirements.
- Developed a Security Managed Service Customer Management System using Flask, streamlining operational workflows for external security clients.

PROJECTS

AI-CERT | *Lead Developer* **Dec. 2025 - Present**

- Engineered a FastAPI relay server that ingests Splunk Enterprise Security notable-event webhooks in real time and routes them to an on-premises LLM through a LiteLLM gateway, which returns a verdict and rationale for each event.
- Enriched LLM prompts with multi-source context — blocklist history, prior detection records, raw logs with packet payloads from WAF and IDS/IPS, and connection logs from firewalls — to improve analysis accuracy and reduce false-positive escalations.
- Ensured production reliability under load through concurrency control, failure recovery, and connection pooling across the LLM and Splunk API integration.
- Built an IP-level incident dashboard with expandable timelines and one-click analyst actions (email notification, false-positive marking) via custom REST APIs, streamlining triage workflows.

Host Firewall Management & Micro-segmentation System | *Lead Developer* **Aug. 2025 - Dec. 2025**

- Developed a web-based management system by integrating **Flask**, **Ansible**, and **nftables** to enforce host-level micro-segmentation within a Zero Trust testbed environment.
- Automated security policy distribution and host-based access control, ensuring consistent and granular security postures across isolated Linux servers.
- Integrated host servers with Splunk via Syslog to enable centralized log retrieval, firewall policy auditing, and login audit trail storage — establishing end-to-end observability within the Zero Trust testbed.

Enterprise IT Asset Management & Rogue Device Detection System | *Lead Developer* **Sep. 2024 - Present**

- Engineered a centralized security platform using **Flask** and the **Splunk Python SDK** to automate the auditing and tracking of enterprise IT assets.
- Developed custom Python scripts to programmatically retrieve and analyze real-time network logs from Splunk, identifying discrepancies between registered assets and active network devices.
- Implemented automated detection logic to identify unauthorized devices within the financial network, significantly enhancing asset visibility and security compliance.

CERTIFICATIONS

CISSP (Certified Information Systems Security Professional) | ISC² **Mar. 2024 - Present**

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, Java, Shell Scripting
Data & Monitoring: ELK Stack (Elasticsearch, Logstash, Kibana), Grafana, MySQL
Security & Networking: SIEM (Splunk), WAF, Anti-DDoS, IPS, IDS, IPSec VPN
Tools and Frameworks: FastAPI, Flask, Django, Node.js, Vue.js, Nginx, Ansible, Git
ML / Data Science: LLM integration via LiteLLM, Splunk MLTK (unsupervised anomaly detection), MATLAB

ADDITIONAL INFORMATION

Volunteer Leadership: World Friends Korea IT Volunteer Corps, Team Leader **Jul. 2017 - Aug. 2017**

- Led a 4-member team, dispatched under Korea's National Information Society Agency, delivering digital literacy instruction (OS, internet, MS Office) to children with autism at SERRIC, a special-education institution under Sri Lanka's Ministry of Defence.

Military Service: Republic of Korea Air Force (ROKAF), Sergeant **Jan. 2015 - Jan. 2017**